

Mixed Tate motives over number fields

(for an algebraic topologist)

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Joint work with Alexander Kupers and Daniil Rudenko

Introduction

The dream

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Q2: What's the "meaning" of their generators and relations?

$F = \text{number field}$ then Borel computed $K_{2i}(F)_{\mathbb{Q}} = 0$ ($i > 0$) and

$$\dim_{\mathbb{Q}}(K_{2i-1}(F)_{\mathbb{Q}}) = \begin{cases} r_2 & \text{if } i \text{ even} \\ r_1 + r_2 & \text{if } i \geq 3 \text{ odd.} \end{cases}.$$

However: what are the generators?

Topological approach to compute K -groups

$K(F)$ = group completion of $\mathbf{BGL}(F) := \bigsqcup_n BGL_n(F) \in \mathcal{A}lg_{E_\infty}(\mathcal{S})$.

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This is just a graded cdga over $\mathbb{Q}$.

Homology: $H_{n,d}(\mathbf{BGL}(F)_\mathbb{Q}) := H_d(\mathbf{BGL}(F)_\mathbb{Q}(n)) = H_d(GL_n(F), \mathbb{Q})$.

Program I

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Step 2. Take E_∞ -homology, $H_{n,d}^{E_\infty}(\mathbf{BGL}(F)_{\mathbb{Q}})$ = “d-cells in rank n in the minimal model for the cdga $\mathbf{BGL}(F)_{\mathbb{Q}}$.”

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- There is a canonical map $H_{n,d}(\mathbf{BGL}(F)_\mathbb{Q}) \rightarrow H_{n,d}^{E_\infty}(\mathbf{BGL}(F)_\mathbb{Q})$.
- There is a *group completion spectral sequence*

$$E_{n,d}^1 = H_{n,d}^{E_\infty}(\mathbf{BGL}(F)_\mathbb{Q}) \Rightarrow K_d(F)_\mathbb{Q},$$

where d^r has bidegree $(-r, -1)$.

Recovers Rognes' rank spectral sequence.

Step 4. Computing E_∞ homology:

$$H_{n,d}^{E_\infty}(\mathbf{BGL}(F)_\mathbb{Q}) = H_{d-2n+2}(GL_n(F), St_n^\infty(F)_\mathbb{Q}),$$

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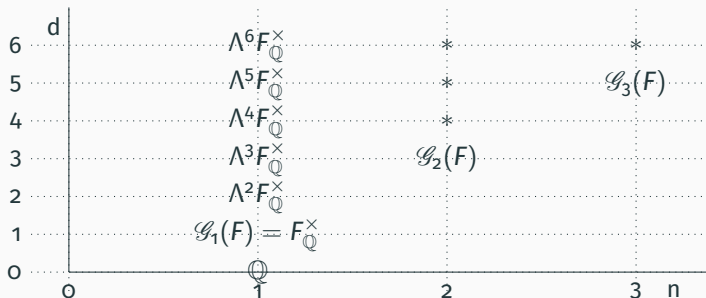
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Corollary: $H_{n,d}^{E_\infty}(\mathbf{BGL}(F)_\mathbb{Q}) = 0$ for $d < 2n - 2$.

In fact, $H_{n,2n-2}^{E_\infty}(\mathbf{BGL}(F)_\mathbb{Q}) = H_0(GL_n(F), St_n^\infty(F)_\mathbb{Q}) = 0$ for $n \geq 2$ too.

The spectral sequence I

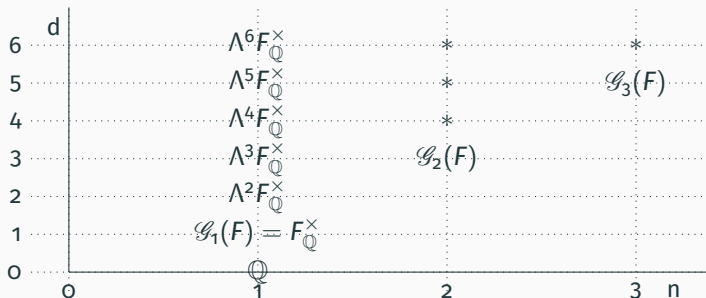
Group completion spectral sequence



where $\mathcal{G}_n(F) := H_{n,2n-1}^{E_{\infty}}(\mathbf{BGL}(F)_{\mathbb{Q}}) = H_1(GL_n(F), St_n^{\infty}(F)_{\mathbb{Q}})$.

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$K_d(F)_{\mathbb{Q}}$ computed on d -th row.

Edge takes form $K_{2n-1}(F)_{\mathbb{Q}} \rightarrow \mathcal{G}_n(F)$.

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$$B_2(F) := \frac{\mathbb{Q}\langle \{x\} : x \in F^\times \setminus \{1\} \rangle}{(\{x\} + \{y\} - \{xy\} - \{\frac{x(1-y)}{1-xy}\} - \{\frac{y(1-x)}{1-xy}\})}$$

Moreover, $d^1 : B_2(F) \rightarrow \Lambda^2 F_{\mathbb{Q}}^\times$ given by

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New observation: $\mathcal{G}(F) := \bigoplus_n \mathcal{G}_n(F)$ form a (graded) Lie coalgebra by Koszul duality between E_∞ and coLie .

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We call $\mathcal{G}(F)$ the *Goncharov Lie coalgebra* of F .

Main results

The main theorem

Theorem (Kupers–Rudenko–S.)

Let F be a number field, then **the Goncharov Lie coalgebra of F is isomorphic to the motivic Lie coalgebra of F** (in a preferred way).

Thus, there is an equivalence of categories

$$\mathrm{coMod}_{\mathcal{G}(F)}^{\mathrm{f},\mathrm{d}}(\mathrm{Vect}_{\mathbb{Q}}^{\mathbb{N}}) \xrightarrow{\cong} \mathrm{MTM}_{\mathbb{Q}}(F).$$

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We also found explicit generators and relations for $\mathcal{G}(F)$ and computed the cobracket on the generators for any field F .

- Using group completion spectral sequence get presentations for $K_4(F)_{\mathbb{Q}}$ and part of $K_5^{(3)}(F)_{\mathbb{Q}}$ for any F .

Consequences

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- Generators of $\mathcal{G}(F) \iff$ Goncharov's motivic correlators in the motivic Lie coalgebra $\Rightarrow$ motivic correlators generate motivic Lie coalgebra for number fields.

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- Conjectural description of full symbolic description of $K_*(F)_{\mathbb{Q}}$ for any F (known for number fields).

K -theory and motivic Lie coalgebra

Mixed Tate motives

Theorem (Beilinson, Deligne, Levine, Goncharov, Voevodsky,...)

Let $F =$ number field. Then $\exists$ a Tannakian category $\text{MTM}_{\mathbb{Q}}(F)$ of Mixed Tate motives over F .

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- “Arbitrary objects are iterated extensions of pure Tate objects.”

E.g. For $x \in F^{\times} \setminus \{1\}$, $H^1(\mathbb{A}^1 \setminus \{0\}, \{1, x\})$ fits into an extension

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- “Extension data is K -theory of F ”

$$\text{Ext}_{\text{MTM}_{\mathbb{Q}}(F)}^i(\mathbb{Q}(-n), \mathbb{Q}(0)) \cong K_{2n-i}^{(n)}(F)_{\mathbb{Q}},$$

the n -th Adams eigenspace of $K_{2n-i}(F)_{\mathbb{Q}}$.

E.g. above extension corresponds to $[x] \in K_1(F)_{\mathbb{Q}}!$

The motivic Lie coalgebra

Tannakian formalism:

$$\mathrm{MTM}_{\mathbb{Q}}(F) \simeq \mathrm{coMod}_{\mathcal{L}^{\mathrm{MTM}}(F)}^{\mathrm{f.d.}}(\mathrm{Vect}_{\mathbb{Q}}^{\mathbb{N}})$$

for a graded Lie coalgebra $\mathcal{L}^{\mathrm{MTM}}(F)$ called the *motivic Lie coalgebra*.

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Then,

$$K_{2n-i}^{(n)}(F)_{\mathbb{Q}} \cong \mathrm{Ext}_{\mathrm{MTM}_{\mathbb{Q}}(F)}^i(\mathbb{Q}(-n), \mathbb{Q}(0)) \cong H_i^{\mathrm{CE}}(\mathcal{L}^{\mathrm{MTM}}(F))_n,$$

the degree n part of the i -th Chevalley-Eilenberg homology of $\mathcal{L}^{\mathrm{MTM}}(F)$:

$$\mathcal{L}^{\mathrm{MTM}}(F) \xrightarrow{\delta} \wedge^2 \mathcal{L}^{\mathrm{MTM}}(F) \xrightarrow{\delta} \wedge^3 \mathcal{L}^{\mathrm{MTM}}(F) \rightarrow \dots$$

“K-theory= Extensions = CE homology of a Lie coalgebra”

Key ingredients of the proof

General fields

Recall that $\mathcal{G}_n(F) = H_1(\mathrm{GL}_n(F), \mathrm{St}_n^\infty(F)_{\mathbb{Q}})$.

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Number fields

- Borel–Yang computed $H_{*,*}(\mathbf{BGL}(F)_\mathbb{Q})$. Using E_∞ -algebra tools one finds $\mathcal{G}(F) \cong \mathrm{cofree}_{\mathrm{coLie}}\left(\bigoplus_n K_{2n-1}(F)_\mathbb{Q}\right)$ *non canonically*.
- Borel's computation $\Rightarrow \mathcal{G}(F)$ and $\mathcal{L}^{\mathrm{MTM}}(F)$ are *abstractly* isomorphic.

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- Universal computation: prove that $\bar{c}_{n,2n-1} \in H_{\text{cts}}^{2n-1}(\text{GL}_n(\mathbb{C}), \mathbb{R})$ factors through $\mathcal{G}_n(\mathbb{C})$: “Borel classes have a polylogarithmic formula”.

Thank you for your attention

Generators of $\mathcal{G}_n(F)$:

$$\{\text{Cor}^{\mathcal{G}}(x_0, \dots, x_n) : x_0, \dots, x_n \in F \text{ not all equal}\}$$

Decomposition relations:

$$\begin{aligned} & \text{Cor}^{\mathcal{G}}(x_0, \dots, x_n) - \text{Cor}^{\mathcal{G}}(y_0, \dots, y_n) \\ &= \sum_{\iota = (\{i_1, j_1\}, \dots, \{i_n, j_n\}) \in T(n)} \text{sign}(\iota) \text{Cor}^{\mathcal{G}}\left(0, \frac{x_{i_1} - x_{j_1}}{y_{i_1} - y_{j_1}}, \dots, \frac{x_{i_n} - x_{j_n}}{y_{i_n} - y_{j_n}}\right), \end{aligned}$$

where we omit terms with $y_{i_k} = y_{j_k}$ for some k and $T(n)$ is a certain set of trees with ordered edges.

Cobacket formula for $x_0, \dots, x_n$ distinct:

$$\delta(\text{Cor}^{\mathcal{G}}(x_0, x_1, \dots, x_n)) = \sum_{j=0}^n \sum_{i=1}^{n-1} \text{Cor}^{\mathcal{G}}(x_j, x_{j+1}, \dots, x_{j+i}) \wedge \text{Cor}^{\mathcal{G}}(x_j, x_{j+i+1}, \dots, x_{j+n})$$